

Maharashtra Public Service Commission Assessment

Subject Name :	Assistant Professor Information Technology Group A and Technical Education Department
Duration :	60
Number of Questions :	100
Total Marks :	200
Display Marks:	Yes
Maximum Instruction Time :	0
Is Section Default? :	No

Question Number : 1 Question Id : 4410091448152 Question Type : MCQ

In a design with precise interrupts, which must hold at an interrupt boundary for the instruction cycle?

Options :

1. IR contains the next instruction to execute and PC points to the current instruction
2. Architectural state reflects completion of all instructions up to and including the faulting instruction
3. Architectural state reflects completion of all instructions strictly before the faulting instruction, with the PC pointing to the faulting instruction
4. Architectural state reflects completion of a subset of earlier instructions depending on pipeline depth

Question Number : 2 Question Id : 4410092105206 Question Type : MCQ

Consider an instruction written in assembly language:

ADD R1, R2

Which statement is correct regarding the given instruction?

Options :

1. ADD is the operand and R1, R2 are opcodes
2. R1 is the opcode and R2 is the operand
3. R1 and R2 are both opcodes
4. ADD is the opcode, while R1 and R2 are operands

Question Number : 3 Question Id : 4410092105261 Question Type : MCQ

Which stage of the instruction cycle is responsible for decoding the operation specified by a machine instruction before execution?

Options :

1. Write-back stage
2. Decode stage
3. Execute stage
4. Fetch stage

Question Number : 4 Question Id : 4410092105262 Question Type : MCQ

Which statement best describes the role of the system bus in the functioning of the Arithmetic Logic Unit (ALU) within a computer's data path?

Options :

1. Manages memory allocation for the ALU & Complete Computer System.
2. Executes arithmetic operations within the ALU & decode instructions.
3. Transfers data between the ALU and other components.
4. Directly controls the ALU's operation modes to execute instructions.

Question Number : 5 Question Id : 4410092105263 Question Type : MCQ

Which design aspect is critical for efficient main memory connection to the CPU in a typical memory hierarchy?

Options :

1. Cache hit rate
2. Main memory access time
3. Virtual memory management
4. CPU-memory bus width

Question Number : 6 Question Id : 4410091378529 Question Type : MCQ

What does the instruction STA 1523A do in the machine instructions?

Options :

1. It stores the contents of the accumulator register into the memory address 1523A.
2. It deletes the contents of the accumulator register into the memory address 1523A.
3. It expunges the contents of the accumulator register into the memory address 1523A.
4. It obliterates the contents of the accumulator register into the memory address 1523A.

Question Number : 7 Question Id : 4410091378654 Question Type : MCQ

What do you mean by SKO with respect to Input-Output instructions?

Options :

1. Skip on Output flag
2. Set Known Output flag
3. Synchronize known Input flag
4. Set known Immediate flag

Question Number : 8 Question Id : 4410091378927 Question Type : MCQ

Among the following, which suits best for the cycle stealing mode?

Options :

1. Bulk data transfers such as loading a program into memory.
2. CPU performance is critical and delays in data transfer are acceptable.
3. Small, frequent data transfers where CPU idle time must be minimized.
4. Bulk data transfers such as storing and saving a program into memory.

Question Number : 9 Question Id : 4410092105207 Question Type : MCQ

Consider the following assembly language code:

CMP R1, R2

JZ TARGET

Which condition must be satisfied for the JUMP to occur?

Options :

1. $R1 < R2$
2. $R1 > R2$
3. $R1 = R2$
4. $R1 \neq R2$

Question Number : 10 Question Id : 4410092105264 Question Type : MCQ

A high-performance system faces CPU bottlenecks during disk-to-memory data transfers. Which DMA feature would best reduce CPU load while maintaining system responsiveness?

Options :

1. DMA system supporting only burst mode transfers
2. Simple DMA controller without cycle stealing
3. DMA with cycle stealing for interleaved transfer
4. Interrupt-driven I/O for all transfers

Question Number : 11 Question Id : 4410091448210 Question Type : MCQ

In a successful synchronous ReadFile on a regular disk file, which statement about the file pointer position is correct

Options :

1. The file pointer advances by the exact number of bytes actually read, which may be less than requested near EOF
2. The file pointer always advances by the number of bytes requested, regardless of bytes actually read
3. The file pointer never advances on read; it only advances on WriteFile
4. The file pointer resets to zero at EOF automatically

Question Number : 12 Question Id : 4410092105090 Question Type : MCQ

Which of the following is a fundamental strategy for deadlock prevention in operating systems?

Options :

1. Ignoring hold and wait
2. Allowing mutual exclusion
3. Denying circular wait conditions
4. Enabling preemption

Question Number : 13 Question Id : 4410092105213 Question Type : MCQ

Which benefit is commonly associated with multithreading in operating system?

Options :

1. Removal of context switching
2. Increased responsiveness
3. Elimination of memory usage
4. Elimination of process creation

Question Number : 14 Question Id : 4410092105393 Question Type : MCQ

A process attempts to read data from a file using the `ReadFile()` system call, but the operation immediately fails with an "invalid handle" error. What is the most likely cause?

Options :

1. The read buffer is smaller than the number of bytes requested.
2. The file pointer has already reached the end of the file.
3. The file handle supplied to `ReadFile()` is invalid.
4. The file was opened with write-only access permissions.

Question Number : 15 Question Id : 4410092105396 Question Type : MCQ

What is a possible drawback of preemptive algorithms compared to non-preemptive ones in terms of system overhead?

Options :

1. They guarantee lower system throughput
2. They may increase context switch frequency
3. They reduce resource utilization
4. They always provide fixed response time

Question Number : 16 Question Id : 4410091448214 Question Type : MCQ

If a process calls CloseHandle on a handle to a process object obtained via OpenProcess while the target process is still running, which statement is correct about the effect on the target process and the handle state?

Options :

1. The target process is terminated immediately and the process object is removed from the system
2. The handle becomes invalid for the caller, but the target process continues running unaffected
3. The handle remains valid until all threads in the caller exit naturally
4. The target process pauses until the handle is reopened by the caller

Question Number : 17 Question Id : 4410092105214 Question Type : MCQ

Suppose process P1 arrives first with a burst time of 20 ms, followed by process P2 with a burst time of 3 ms and finally process P3 arrives with a burst time of 1 ms. Under FCFS scheduling, which process will have the largest waiting time?

Options :

1. Both P2 and P3
2. P1
3. P2
4. P3

Question Number : 18 Question Id : 4410092105267 Question Type : MCQ

A system administrator is analyzing a process scheduling log and notices that Process A arrived at time 2 ms, Process B at time 5 ms, and Process C at time 8 ms. If a fourth process, Process D, is scheduled to arrive just after Process B but before Process C, which arrival time would be most consistent with the pattern and ensure no overlap with other processes?

Options :

1. 6 ms
2. 10 ms
3. 8 ms
4. 4 ms

Question Number : 19 Question Id : 4410092105395 Question Type : MCQ

A developer needs lightweight thread creation and switching for a scientific simulation. Which approach best achieves this with minimal operating system interference?

Options :

1. Delegating thread control to a system-wide daemon process.
2. Using kernel-managed threads with direct OS scheduling.
3. Implementing threads entirely within the application using a thread library.
4. Employing hybrid threading with mixed management.

Question Number : 20 Question Id : 4410092105397 Question Type : MCQ

A process terminates in a dynamic partitioning system, releasing 150 KB between two occupied blocks. What is the immediate effect on the memory map?

Options :

1. A single 150 KB free space appears between two allocations
2. The adjacent occupied blocks are merged into a larger allocation
3. Two separate free spaces are created at different locations
4. Non-adjacent free spaces are automatically combined

Question Number : 21 Question Id : 4410091466960 Question Type : MCQ

What is 2's complement used for?

Options :

1. Representing signed integers and subtraction
2. Only positive numbers
3. Floating-point
4. ASCII characters

Question Number : 22 Question Id : 4410092105271 Question Type : MCQ

Which of the following logic gates outputs a HIGH signal if at least one input is HIGH?

Options :

1. XOR gate
2. NAND gate
3. OR gate
4. AND gate

Question Number : 23 Question Id : 4410091456551 Question Type : MCQ

If the binary number 10011011.01 is divided by 2^4 , the correct resulting binary is:

Options :

1. 00001001.1011
2. 1001.101101
3. 1001.1011011
4. 1001.1011

Question Number : 24 Question Id : 4410091456557 Question Type : MCQ

In a fixed-point representation with a signed 8-bit number where 4 bits are allocated to the fractional part, what is the decimal value represented by the binary number 11111000?

Options :

1. -0.5
2. -1.0
3. -8.0
4. -0.25

Question Number : 25 Question Id : 4410091467134 Question Type : MCQ

Why is NOR universal?

Options :

1. Requires NAND
2. Limited to inversion
3. Can implement any boolean function using NOR alone
4. Only for AND/OR

Question Number : 26 Question Id : 4410091356308 Question Type : MCQ

What is the output of a **D flip-flop** after a clock pulse if $D = 1$ and the previous $Q = 0$?

Options :

1. 0
2. 1
3. Depends on the previous state
4. Cannot be determined

Question Number : 27 Question Id : 4410092105092 Question Type : MCQ

A digital system receives a signed 8-bit binary number in two's complement form. An error occurs, and the number is interpreted as unsigned binary. What is the most likely impact on the system's output if the original number was negative?

Options :

1. The output will represent a small positive value close to zero.
2. The output will represent a large positive value than the intended negative value.
3. The output will become zero regardless of input.
4. The output will invert the value to a negative number.

Question Number : 28 Question Id : 4410092105093 Question Type : MCQ

A security system requires a logic circuit to trigger an alarm only when not all of the sensors are active. Which logic gate would best fulfill this requirement if each sensor provides a HIGH signal when active?

Options :

1. AND gate
2. NAND gate
3. NOR gate
4. OR gate

Question Number : 29 Question Id : 4410092105272 Question Type : MCQ

A digital system requires a flip-flop that toggles its output when both inputs are high and holds its state when both inputs are low. Which flip-flop should you use, and what modifications, if any, would you apply to convert a JK flip-flop into this required flip-flop?

Options :

1. Use a JK flip-flop without any modification
2. Use a JK flip-flop with J and K connected together
3. Use an SR flip-flop with both inputs tied low
4. Use a D flip-flop with input connected to output

Question Number : 30 Question Id : 4410092105398 Question Type : MCQ

A memory address is represented as 9C7F in hexadecimal. What is the sum of this address and the hexadecimal value 3A81?

Options :

1. D600
2. D780
3. D700
4. C700

Question Number : 31 Question Id : 4410091448280 Question Type : MCQ

A production table Orders(id INT PRIMARY KEY, created_at DATETIME NOT NULL, note VARCHAR(50) NULL) has tens of millions of rows. You must make note NOT NULL and default to an empty string without causing application inserts to fail midway. Which statement most safely enforces this in engines that backfill existing NULLs?

Options :

1. ALTER TABLE Orders MODIFY note VARCHAR(50) NOT NULL DEFAULT "";
2. ALTER TABLE Orders ADD CONSTRAINT df_note DEFAULT "" ON COLUMN note;
3. ALTER TABLE Orders ALTER COLUMN note SET DEFAULT "" NOT NULL IF NULL;
4. ALTER TABLE Orders SET COLUMN note VARCHAR(50) DEFAULT "" NOT NULL USING "";

Question Number : 32 Question Id : 4410091449466 Question Type : MCQ

In an ER diagram, an attribute name that is underlined with a dashed line, as shown below, represents which of the following?

Options :

1. A primary key attribute of a strong entity
2. A derived attribute of a weak entity
3. A multivalued attribute of an entity
4. A discriminating attribute of a weak entity set

Question Number : 33 Question Id : 4410092105094 Question Type : MCQ

Which symbol is typically used in ER diagrams to represent a relationship between two entities?

Options :

1. Double line
2. Diamond
3. Rectangle
4. Ellipse

Question Number : 34 Question Id : 4410092105220 Question Type : MCQ

What is the effect of the following SQL command in Database Management System?

DELETE FROM Employee;

Options :

1. Deletes the employee table structure
2. Deletes only one row from Employee table
3. Deletes all rows from Employee table
4. Deletes only the rows that contain NULL values

Question Number : 35 Question Id : 4410092105402 Question Type : MCQ

Suppose you execute DROP VIEW statement on a view that is referenced by stored procedures. What is the likely consequence?

Options :

1. All references are updated to point elsewhere automatically
2. Stored procedures that reference the dropped view will fail upon execution
3. Procedures begin referencing a different object of similar name
4. A replacement view is automatically created

Question Number : 36 Question Id : 4410091491703 Question Type : MCQ

In the schedule of transactions, which of the following READ operations may lead to a dirty read?

Options :

1. Reading uncommitted data written by another transaction.
2. Reading only committed data.
3. Reading from a read-only view.
4. Reading from temporary local variables.

Question Number : 37 Question Id : 4410092105273 Question Type : MCQ

A company wants to design a new relational database to handle employee records and project assignments. Considering the concept of tuples in the relational model, which approach best ensures that each employee-project assignment is uniquely and efficiently represented?

Options :

1. Create a relation where each tuple combines one employee with one project assigned
2. Use a single tuple per project listing all assigned employees in a multi-valued attribute
3. Include multiple project attributes within a single employee tuple
4. Store all project assignments for an employee in a single tuple as a comma-separated list

Question Number : 38 Question Id : 4410092105277 Question Type : MCQ

A banking application is processing a fund transfer transaction. Just before the COMMIT command is issued, a power failure occurs, and the database system restarts. Which of the following outcomes best reflects proper transaction processing in this situation?

Options :

1. The transaction will be rolled back, and no account changes will occur
2. The system will automatically retry the transaction upon restart.
3. The updates will be permanently reflected, even though COMMIT was not issued.
4. Only one account will reflect the new balance, leading to inconsistency.

Question Number : 39 Question Id : 4410092105401 Question Type : MCQ

A database schema includes a relation $R(A, B, C)$ and a current instance S . Which operation could produce a new S that is not a valid relation instance?

Options :

1. Adding a tuple to S that is identical to an existing tuple in all attributes
2. Renaming attribute B to D in all tuples of S
3. Deleting a tuple from S based on a condition
4. Permuting the columns of S without changing values

Question Number : 40 Question Id : 4410092105403 Question Type : MCQ

Consider a relation $R(U, V, W, X)$ with the functional dependencies:

- $UW \rightarrow X$
- $X \rightarrow V$
- $V \rightarrow U$

If UW is the candidate key of the relation, what is the highest normal form satisfied by R ?

Options :

1. The relation is in BCNF.
2. The relation is in 2NF but not 3NF.
3. The relation is in 3NF but not BCNF.
4. The relation is only in 1NF.

Question Number : 41 Question Id : 4410092105225 Question Type : MCQ

Wi-Fi technology is based on which IEEE standard?

Options :

1. IEEE 802.11
2. IEEE 802.16
3. IEEE 802.15
4. IEEE 802.3

Question Number : 42 Question Id : 4410092105100 Question Type : MCQ

Which statement best describes a key difference between circuit switching and virtual circuit switching in computer networks?

Options :

1. Circuit switching uses packet-based transmission, while virtual circuit switching sends data as a continuous stream only.
2. Circuit switching establishes a dedicated physical path for the entire session, while virtual circuit switching uses logical paths that may share physical resources with other connections.
3. Both circuit switching and virtual circuit switching require a dedicated physical path for each session without sharing resources.
4. Virtual circuit switching does not maintain any end-to-end path, while circuit switching uses multiple dynamic routes per packet.

Question Number : 43 Question Id : 4410091445230 Question Type : MCQ

How does HTTP/2 improve over HTTP/1.1?

Options :

1. By eliminating caching
2. By requiring multiple connections
3. By multiplexing requests over a single TCP connection
4. By using text-based headers

Question Number : 44 Question Id : 4410092105279 Question Type : MCQ

An organization wants to implement a secure and scalable electronic mail system to support remote employees and seamless communication with external partners. Which of the following architectural approaches best addresses the requirements, considering common application layer services and integration with authentication mechanisms?

Options :

1. Configuring multiple POP3 servers without centralized authentication
2. Deploying a distributed mail server with SMTP, IMAP, and LDAP-based authentication
3. Using IMAP servers with basic password authentication only, without protocol redundancy
4. Implementing a single centralized mail server using only SMTP with local user accounts

Question Number : 45 Question Id : 4410092105405 Question Type : MCQ

A system designer must implement selective retransmission of corrupted segments without affecting flow control mechanisms. At which OSI layer should this functionality be embedded, and why?

Options :

1. Selective retransmission should be implemented at the Data Link layer for local error correction
2. Selective retransmission should be implemented at the Transport layer for end-to-end reliability
3. Selective retransmission should be implemented at the Network layer for optimal routing
4. Selective retransmission should be implemented at the Application layer for user transparency

Question Number : 46 Question Id : 4410092105111 Question Type : MCQ

Which searching algorithm checks each element in a list sequentially until the desired value is found or the list ends?

Options :

1. Binary Search
2. Linear Search
3. Jump Search
4. Interpolation Search

Question Number : 47 Question Id : 4410092105231 Question Type : MCQ

What is the time complexity of Binary search algorithm in the best case?

Options :

1. $O(\log n)$
2. $O(1)$
3. $O(n)$
4. $O(2*n)$

Question Number : 48 Question Id : 4410092105411 Question Type : MCQ

When analyzing the memory required by an algorithm, what should be excluded from the calculation of auxiliary space?

Options :

1. Temporary variables created during execution
2. Space used for recursive call stack
3. Space used to store input data
4. Additional data structures built during computation

Question Number : 49 Question Id : 4410092105234 Question Type : MCQ

Consider the array of integers: [8, 4, 6, 2, 7].

What will be the contents of the array after the third pass of Insertion Sort?

Options :

1. 4, 8, 6, 2, 7
2. 2, 4, 6, 8, 7
3. 4, 6, 8, 2, 7
4. 2, 4, 8, 6, 7

Question Number : 50 Question Id : 4410092105281 Question Type : MCQ

Which of the following best describes an algorithm in the context of Design and Analysis of Algorithms?

Options :

1. A random list of instructions to perform operations
2. A mathematical proof for correctness
3. A finite sequence of defined steps to solve a problem
4. A set of hardware components for computation

Question Number : 51 Question Id : 4410092105283 Question Type : MCQ

Which of the following best describes the worst-case time complexity analysis of an algorithm?

Options :

1. It estimates the minimum time taken by an algorithm for any input size
2. It estimates the maximum time taken by an algorithm for any input size
3. It estimates the average time taken over all possible inputs of a given size
4. It estimates the time taken under typical and practical input conditions

Question Number : 52 Question Id : 4410092105284 Question Type : MCQ

Which of the following statements correctly compares the time complexity of searching for an element in a sorted array using binary search to linear search?

Options :

1. Binary search has logarithmic time complexity, while linear search has linear time complexity
2. Both binary search and linear search have quadratic time complexity in all cases
3. Both binary search and linear search have constant time complexity in all cases
4. Binary search has linear time complexity, while linear search has logarithmic time complexity

Question Number : 53 Question Id : 4410092105415 Question Type : MCQ

Suppose an array of length eight is already sorted in descending order. How many total comparisons will selection sort perform to fully sort it in ascending order?

Options :

1. 28
2. 16
3. 20
4. 32

Question Number : 54 Question Id : 4410092105106 Question Type : MCQ

A software engineer is evaluating the performance of a dynamic array that occasionally requires resizing when new elements are inserted. Although some insertions take significantly longer due to resizing, the average cost per insertion remains low over a large sequence of operations. Which concept best explains this behavior?

Options :

1. Asymptotic analysis ignores the cost of expensive operations such as resizing.
2. Amortized analysis evaluates the average cost of a sequence of operations, guaranteeing average performance even in the worst-case sequence.
3. Best-case analysis assumes that resizing never occurs during execution.
4. Worst-case analysis measures the maximum cost of every individual operation.

Question Number : 55 Question Id : 4410092105228 Question Type : MCQ

Which of the following errors can be identified by testing an algorithm with unusual or extreme input values?

Options :

1. Compilation errors
2. Boundary-value errors
3. Formatting errors
4. Linking errors

Question Number : 56 Question Id : 4410092105229 Question Type : MCQ

Which algorithm generally requires the least additional memory to solve a problem of size 'n'?

Options :

1. Recursive algorithm with deep recursion
2. Algorithm using dynamic programming tables
3. Algorithm using multiple auxiliary arrays
4. An in-place sorting algorithm

Question Number : 57 Question Id : 4410092105233 Question Type : MCQ

Consider the following array of integers: [12, -5, 8, 12, -9, 3, 8, -1, 15, -5]

The array is sorted in ascending order using Selection Sort.

What will be the contents of the array after the second pass of Selection Sort?

Options :

1. -9, -5, -1, 12, 12, 3, 8, 8, 15, -5
2. -9, -5, 8, -5, 12, 3, 8, -1, 15, 12
3. -9, -1, 8, 12, 12, 3, 8, -5, 15, -5
4. -9, -5, 8, 12, 12, 3, 8, -1, 15, -5

Question Number : 58 Question Id : 4410092105290 Question Type : MCQ

A database contains millions of sorted records, but each record's size varies considerably. You are tasked with designing a search system that minimizes disk access latency using binary search. Which design choice best leverages binary search for optimal performance in this scenario?

Options :

1. Sequentially scan records until the desired one is found
2. Sort records by their size first, then use binary search based on size
3. Apply binary search directly to the variable-sized records in storage
4. Create an index table with fixed-size record pointers before binary search

Question Number : 59 Question Id : 4410092105416 Question Type : MCQ

Consider Bubble Sort with a modification: compare and swap every k -th pair of adjacent elements in each pass ($k > 1$). What is a possible consequence of this change?

Options :

1. The modification has no effect on the output for any input.
2. The sorting process becomes faster without affecting correctness.
3. The overall time complexity of Bubble Sort is reduced to linear.
4. Some elements may never be compared, resulting in incomplete sorting.

Question Number : 60 Question Id : 4410092105417 Question Type : MCQ

Given a scenario where a linear search is repeatedly applied to very large datasets stored on external memory (e.g., disk), which factor most critically impacts the overall search efficiency compared to in-memory searches?

Options :

1. The use of pointers versus indices in code
2. The speed of the CPU executing the search algorithm
3. The choice of programming language for implementation
4. The time spent accessing each element from disk

Question Number : 61 Question Id : 4410092105240 Question Type : MCQ

Which states are initially separated during Deterministic Finite Automata (DFA) minimization algorithm?

Options :

1. Reachable and unreachable states
2. Start and final states
3. Final and non-final states
4. Equivalent states

Question Number : 62 Question Id : 4410092105291 Question Type : MCQ

Which of the following best defines a Finite State Machine in the context of computation theory?

Options :

1. A model that requires randomness for state transitions
2. A mathematical model with finite states and transitions
3. A structure with unbounded memory for computation
4. A device that operates with infinite states and memory

Question Number : 63 Question Id : 4410092105117 Question Type : MCQ

Which of the following regular expressions correctly defines the set of all strings over the alphabet {a, b} that contain at least one 'a'?

Options :

1. $(b|ab)^*$
2. $(a|b)^*$
3. $b^*ab^*(a|b)^*$
4. a^*b^*

Question Number : 64 Question Id : 4410092105293 Question Type : MCQ

Which of the following problems cannot be solved by any finite state machine, regardless of the number of states?

Options :

1. Recognizing strings containing a specific substring
2. Verifying that a string starts and ends with the same symbol
3. Checking whether a string has equal counts of two input symbols
4. Counting up to a fixed number of occurrences of a symbol

Question Number : 65 Question Id : 4410092105420 Question Type : MCQ

An engineer must design a device that responds differently to the same input based on a sequence of prior actions. Which computational model is most suitable for this application?

Options :

1. Turing machine simulating infinite tape memory
2. Stack automaton with push and pop operations
3. Combinational circuit using Boolean logic
4. Finite state machine tracking historical input sequences

Question Number : 66 Question Id : 4410092105237 Question Type : MCQ

Which component is NOT required by a Deterministic Finite Automaton (DFA)?

Options :

1. Stack memory
2. Set of states
3. Transition function
4. Input alphabet

Question Number : 67 Question Id : 4410092105238 Question Type : MCQ

If a Mealy machine is in state q_0 and receives input 1, producing output 0 and moving to q_1 , then what is the label on the transition from state q_0 to state q_1 ?

Options :

1. 1
2. 0
3. $q_1/0$
4. $1/0$

Question Number : 68 Question Id : 4410092105296 Question Type : MCQ

Suppose you are designing a finite state machine (FSM) to recognize the language defined by the regular expression $(a|b)^*abb$. Which of the following statements best describes the necessary structure of the FSM for this task?

Options :

1. The FSM should have only one state regardless of input since $(a|b)^*$ matches any string.
2. The FSM must reject all strings that contain any occurrence of 'a' or 'b'.
3. The FSM must track the substring 'abb' after any combination of 'a' and 'b'.
4. The FSM only needs to count the total number of 'a's and 'b's in the input.

Question Number : 69 Question Id : 4410092105418 Question Type : MCQ

Consider a language over $\{x,y\}$ where every string has an even number of x's and odd number of y's. What can be said about the minimal DFA recognizing this language?

Options :

1. It requires four distinct states to track parity
2. Three states are sufficient for tracking counts
3. It can be constructed with two states
4. Six states are required for complete tracking

Question Number : 70 Question Id : 4410092105424 Question Type : MCQ

Consider the regular expression $R = (a + b)^*a(a + b)^*b(a + b)^*$, which of the following best describes the language generated by $L(R)$?

Options :

1. All strings over $\{a, b\}$ containing atleast one a before one b
2. All strings over $\{a, b\}$ with only one a and only one b
3. All strings over $\{a, b\}$ where a and b are adjacent to each other
4. All strings over $\{a, b\}$ containing one b before one a

Question Number : 71 Question Id : 4410092105244 Question Type : MCQ

What is the output of the given program in C language?

```
#include <stdio.h>
void fun(int x) {
    x = x + 10;
}
int main() {
    int a = 5;
    fun(a);
    printf("%d", a);
    return 0;
}
```

Options :

1. Compilation Error
2. 10
3. 15
4. 5

Question Number : 72 Question Id : 4410092105298 Question Type : MCQ

Which of the following best describes the primary role of an 'if' statement in programming control flow?

Options :

1. It checks the syntax correctness of code.
2. It executes code regardless of conditions.
3. It loops through a block of code repeatedly.
4. Executes code only when the condition is true.

Question Number : 73 Question Id : 4410092105299 Question Type : MCQ

When a variable is passed to a function using pass by value, which of the following statements best describes what happens inside the function?

Options :

1. The function receives a reference and can directly modify the original variable
2. The function gets a copy and cannot change the original variable
3. The function can access the memory address and alter the original variable
4. The function automatically synchronizes changes back to the original variable

Question Number : 74 Question Id : 4410092105300 Question Type : MCQ

Which of the following best describes the declaration of a two-dimensional integer array with 3 rows and 4 columns in C programming language?

Options :

1. `int arr(3,4);`
2. `int arr[3,4];`
3. `int arr[3][4];`
4. `int arr[4][3];`

Question Number : 75 Question Id : 4410092105429 Question Type : MCQ

In a circular singly linked list, after locating the node to be deleted and its predecessor, how many link updates are required to remove the node from the list?

Options :

1. All pointers in the list must be updated
2. Two updates for both predecessor and successor
3. One update for the predecessor's next pointer
4. No pointer updates are required

Question Number : 76 Question Id : 4410092105122 Question Type : MCQ

A software engineer is optimizing a function that processes a 2D array using pointer arithmetic for maximum efficiency. Given a 2D integer array `arr[4][5]` in C, which pointer expression correctly accesses the element in the third row and fourth column?

Options :

1. `*((*arr) + 8)`
2. `*(*(arr + 3) + 2)`
3. `arr[2][3]`
4. `*(*(arr + 2) + 3)`

Question Number : 77 Question Id : 4410092105123 Question Type : MCQ

A software engineer needs to optimize an algorithm for a system where frequent insertions are performed in the middle of a linked list. Which data structure modification would best minimize the time complexity of these insertions, considering the constraints of a singly linked list?

Options :

1. Increase node size to store more data
2. Maintain a pointer to the node before the desired insertion point
3. Use a dummy head node for all insertions
4. Store the list in an array for direct access

Question Number : 78 Question Id : 4410092105246 Question Type : MCQ

Which deletion operation typically takes $O(1)$ time in an array of size 'n'?

Options :

1. Delete middle element
2. Delete last element
3. Delete all elements one by one
4. Delete first element

Question Number : 79 Question Id : 4410092105426 Question Type : MCQ

In the context of programming languages, a function defined inside another function accesses a variable declared in the outer function. If the outer function returns the inner function, which property is demonstrated?

Options :

1. Variable shadowing in nested function definitions
2. Automatic type inference in inner functions
3. Lexical scoping with persistent variable binding
4. Dynamic scoping based on call stack sequence

Question Number : 80 Question Id : 4410092105430 Question Type : MCQ

Consider the prefix expression $+ * 3 2 / 8 4$. Which infix form is equivalent?

Options :

1. $(3 + 2) * (8 / 4)$
2. $(3 * 2) + (8 / 4)$
3. $3 * (2 + 8) / 4$
4. $3 + (2 * 8) / 4$

Question Number : 81 Question Id : 4410092105254 Question Type : MCQ

Shift-Reduce parsing is an example of which parsing method in compiler design?

Options :

1. Recursive parsing
2. Bottom-up parsing
3. Top-down parsing
4. Semantic parsing

Question Number : 82 Question Id : 4410092105255 Question Type : MCQ

Constant propagation is classified as which type of compiler optimization?

Options :

1. Syntax optimization
2. Lexical optimization
3. Data-flow optimization
4. Machine-dependent optimization

Question Number : 83 Question Id : 4410092105126 Question Type : MCQ

Which of the following best explains the role of compilers in the evolution of programming languages?

Options :

1. Compilers only converted assembly instructions to machine code, limiting language development.
2. Compilers eliminated the need for program optimization by automating all code improvements.
3. Compilers allowed high-level languages to be translated into machine code, enabling portability across different hardware.
4. Compilers replaced the need for assembly language by directly executing source code line by line.

Question Number : 84 Question Id : 4410092105132 Question Type : MCQ

In the context of the constant propagation framework, which statement best describes the role of a transfer function within a basic block?

Options :

1. It resets all variable values to undefined at the block's end.
2. It ignores assignments and passes input information to the output.
3. It updates the constant information for variables based on assignments in the block.
4. It maintains all variable values unchanged throughout the block.

Question Number : 85 Question Id : 4410092105249 Question Type : MCQ

Which language processor translates an entire high-level language program into machine code before execution?

Options :

1. Compiler
2. Loader
3. Interpreter
4. Assembler

Question Number : 86 Question Id : 4410092105250 Question Type : MCQ

Which of the following is NOT an application of Compiler technology?

Options :

1. Text processing and pattern matching tools
2. Database query processing systems
3. Programming languages
4. CPU scheduling algorithms

Question Number : 87 Question Id : 4410092105251 Question Type : MCQ

Which of the following is typically removed by the lexical analyser of compiler design?

Options :

1. Variable declarations
2. Function definitions
3. Parse trees
4. White spaces and comments

Question Number : 88 Question Id : 4410092105253 Question Type : MCQ

LL(1) parsing is an example of which parsing method in compiler design?

Options :

1. Shift-reduce parsing
2. Bottom-up parsing
3. Top-down parsing
4. Operator precedence parsing

Question Number : 89 Question Id : 4410092105310 Question Type : MCQ

Suppose you are designing a constant propagation framework for a compiler. In a program with multiple branches where a variable is assigned different constant values in different branches, which data flow value should be assigned to that variable at the merge point to ensure correctness of the analysis?

Options :

1. Assign the constant value obtained from the second branch only
2. Assign the constant value obtained from the first branch only
3. Assign the uninitialized value ($\perp$) to the variable at the merge point
4. Assign the undefined value (T) to the variable at the merge point

Question Number : 90 Question Id : 4410092105436 Question Type : MCQ

In Syntax Analysis, for the grammar $S \rightarrow AB$, $A \rightarrow \epsilon$ and $B \rightarrow b$, which terminal symbol belongs to FOLLOW(A)?

Options :

1. \$
2. b
3. ϵ
4. a

Question Number : 91 Question Id : 4410092124785 Question Type : MCQ

Which of the following best describes a group in algebraic structures?

Options :

1. A set with an associative binary operation, an identity, and inverses for all elements
2. A set with only an associative binary operation
3. A set with a commutative operation and identity
4. A set with two operations and distributivity

Question Number : 92 Question Id : 4410092124798 Question Type : MCQ

Given the statements: 'If it rains, the ground is wet' and 'If the ground is not wet, then it did not rain,' what logical relationship do these two statements illustrate?

Options :

1. They are inverses of each other.
2. They are contrapositives of each other.
3. They are logically equivalent but not related as converse, inverse, or contrapositive.
4. They are converses of each other.

Question Number : 93 Question Id : 4410092124797 Question Type : MCQ

A transportation network is being modeled as a graph where each city is a vertex, and each direct flight between cities is an edge. The network does not allow flights from a city to itself, nor does it allow multiple flights between the same pair of cities. Which type of graph best models this transportation network?

Options :

1. Pseudograph
2. Simple graph
3. Multigraph
4. Directed graph

Question Number : 94 Question Id : 4410091362142 Question Type : MCQ

The nth derivative of $f(x) = e^{ax}$ is _____

Options :

1. $a^n e^{ax}$
2. $na^{n-1} e^{ax}$
3. ae^{nx}
4. $n!e^{ax}$

Question Number : 95 Question Id : 4410092124813 Question Type : MCQ

A company is designing a rectangular storage container with an open top and a fixed volume. To minimize the material cost, the surface area must be minimized. Given the function $S(x, y) = x*y + 2h(x + y)$, where h is determined by the volume constraint, which method should be used to find the minimum surface area for given volume?

Options :

1. Use the method of Lagrange multipliers to find the constrained minimum
2. Set all first partial derivatives to zero and solve directly
3. Substitute the constraint into the function and maximize
4. Only use partial derivatives with respect to one variable

Question Number : 96 Question Id : 4410092124825 Question Type : MCQ

Which of the following best describes an eigenvector of a square matrix A?

Options :

1. A vector with all positive entries
2. A vector that is mapped to zero by A
3. A vector that is orthogonal to all other vectors
4. A non-zero vector v such that Av equals a scalar multiple of v

Question Number : 97 Question Id : 4410092124823 Question Type : MCQ

Given a 3x3 matrix, which scenario would make the cofactor expansion along the third column the simplest?

Options :

1. The third column contains one zeros and two nonzero element.
2. The third column contains two zeros and one nonzero element.
3. Each entry in the third column is a different nonzero value.
4. The third column contains all identical nonzero values.

Question Number : 98 Question Id : 4410092124831 Question Type : MCQ

A company uses Cramer's Rule to solve a system of three linear equations representing the production rates of three machines. If the determinant of the coefficient matrix is nonzero, but the determinant associated with one variable is zero, what does this imply about the solution for that variable?

Options :

1. The system has no solution
2. The variable is always positive
3. The variable is uniquely zero in the solution
4. The variable has infinitely many possible values

Question Number : 99 Question Id : 4410091557080 Question Type : MCQ

An electronics store wants to determine which type of headphone to restock based on weekly sales. If over four weeks, the sales for models A, B, C, and D were: Week 1 (A: 10, B: 11, C: 8, D: 10), Week 2 (A: 14, B: 10, C: 12, D: 10), Week 3 (A: 10, B: 14, C: 10, D: 8), Week 4 (A: 12, B: 10, C: 10, D: 14), which model should the store prioritize based on the mode of total sales?

Options :

1. Model A
2. Model B
3. Model C
4. Model D

Question Number : 100 Question Id : 4410092124836 Question Type : MCQ

When approximating a binomial variable with parameters $n = 80$ and $p = 0.7$ by a normal variable, what is the variance of the normal approximation?

Options :

1. Variance is 80
2. Variance is 16.8
3. Variance is 56
4. Variance is 24